



↪ Interactive Problems Problem solving

Problem Solving Session

- Gaurish

Problem 1:

<https://codeforces.com/contest/1479/problem/A>

A. Searching Local Minimum

time limit per test: 2 seconds

memory limit per test: 512 megabytes

This is an interactive problem.

Homer likes arrays a lot and he wants to play a game with you.

Homer has hidden from you a permutation $a_1, a_2, \dots, a_n$ of integers 1 to n . You are asked to find any index k ($1 \leq k \leq n$) which is a local minimum.

For an array $a_1, a_2, \dots, a_n$, an index i ($1 \leq i \leq n$) is said to be a *local minimum* if $a_i < \min\{a_{i-1}, a_{i+1}\}$, where $a_0 = a_{n+1} = +\infty$. An array is said to be a permutation of integers 1 to n , if it contains all integers from 1 to n exactly once.

Initially, you are only given the value of n without any other information about this permutation.

At each interactive step, you are allowed to choose any i ($1 \leq i \leq n$) and make a query with it. As a response, you will be given the value of a_i .

You are asked to find any index k which is a local minimum **after at most 100 queries**.

8 1 1 1 1 1 1 1 1 8

$$i \rightarrow q_i < q_{i-1}$$

100 queries $q_i < q_{i+1}$

→ permutation

1st obs

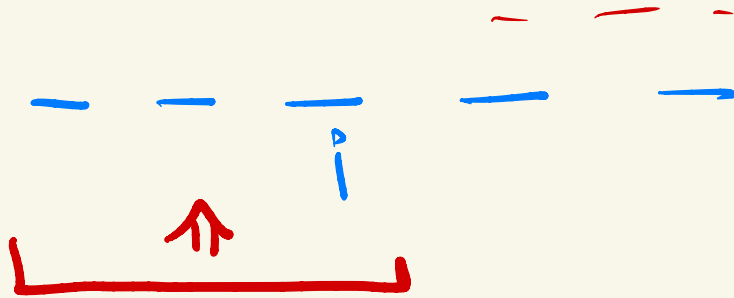
↳ no two elements are the same

$$\underline{i} \quad a_i < a_{i+1}$$

$$a_i > a_{i+1}$$

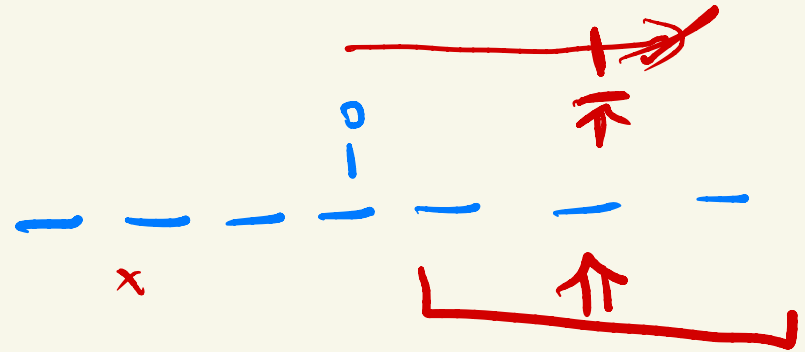
index i

$$a_i < a_{i+1}$$

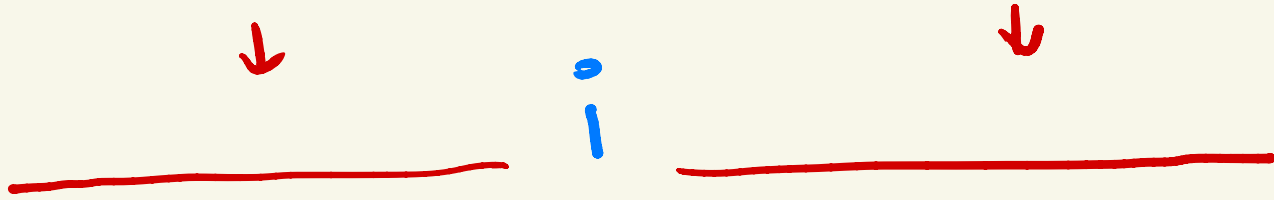


local minimum has to lie near

$$a_i > a_{i+1}$$

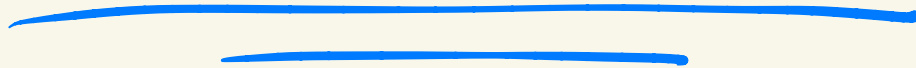


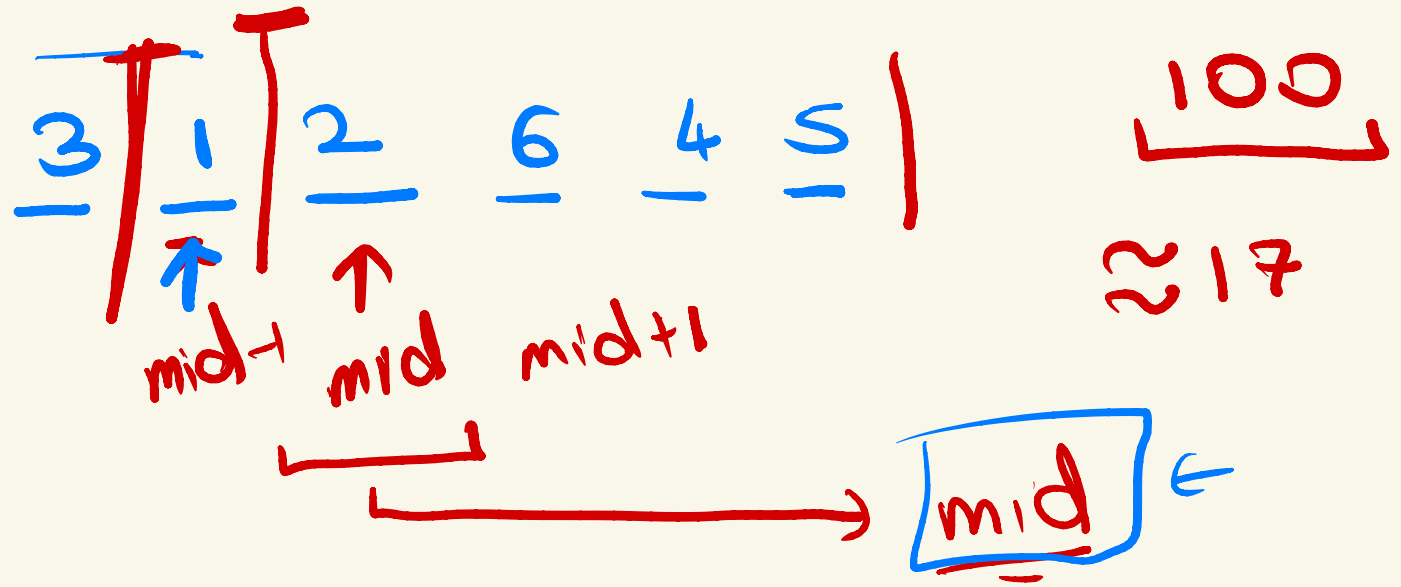
local minimum does lie on this half



Discarding one part of the array

Binary Search





↓

$$\underline{\text{arr}[\text{mid}-1] < \text{arr}[\text{mid}]}$$

or

$$\text{arr}[\text{mid}+1] < \text{arr}[\text{mid}]$$

↗ ↘

Interactive problems

```
→ query function ( L, r ) {  
    → cout << "? L r" << endl;  
    int inp; cin << inp;  
    return  
}
```

Problem 2:

<https://codeforces.com/contest/1934/problem/C>

C. Find a Mine

time limit per test: 2 seconds

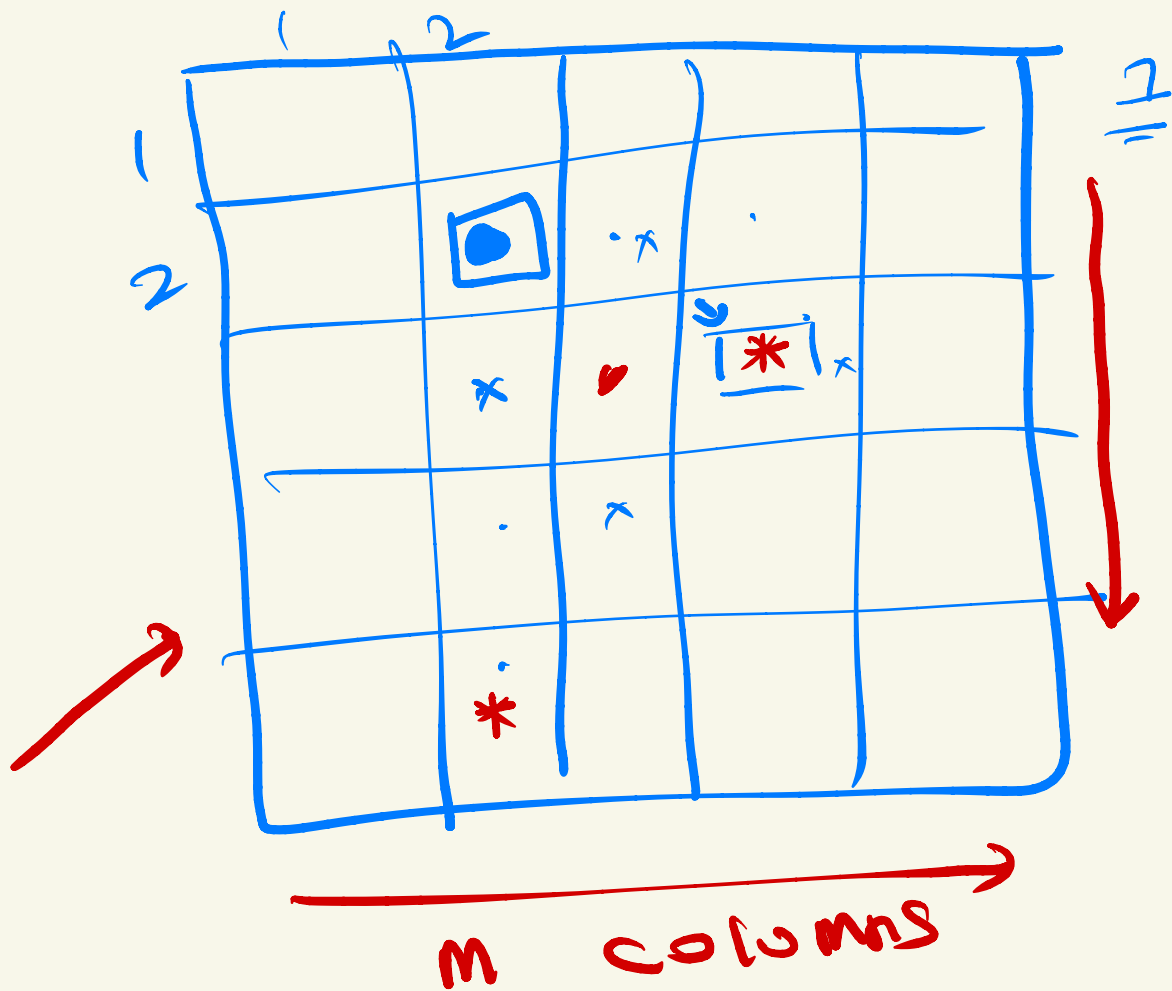
memory limit per test: 256 megabytes

This is an interactive problem.

→ You are given a grid with n rows and m columns. The coordinates (x, y) represent the cell on the grid, where x ($1 \leq x \leq n$) is the row number counting from the top and y ($1 \leq y \leq m$) is the column number counting from the left. It is guaranteed that there are exactly 2 mines in the grid at **distinct** cells, denoted as (x_1, y_1) and (x_2, y_2) . You are allowed to make no more than 4 queries to the interactor, and after these queries, you need to provide the location of **one of the mines**.

In each query, you can choose any grid cell (x, y) , and in return, you will receive the minimum Manhattan distance from both the mines to the chosen cell, i.e., you will receive the value $\min(|x - x_1| + |y - y_1|, |x - x_2| + |y - y_2|)$.

Your task is to determine the location of one of the mines after making the queries.



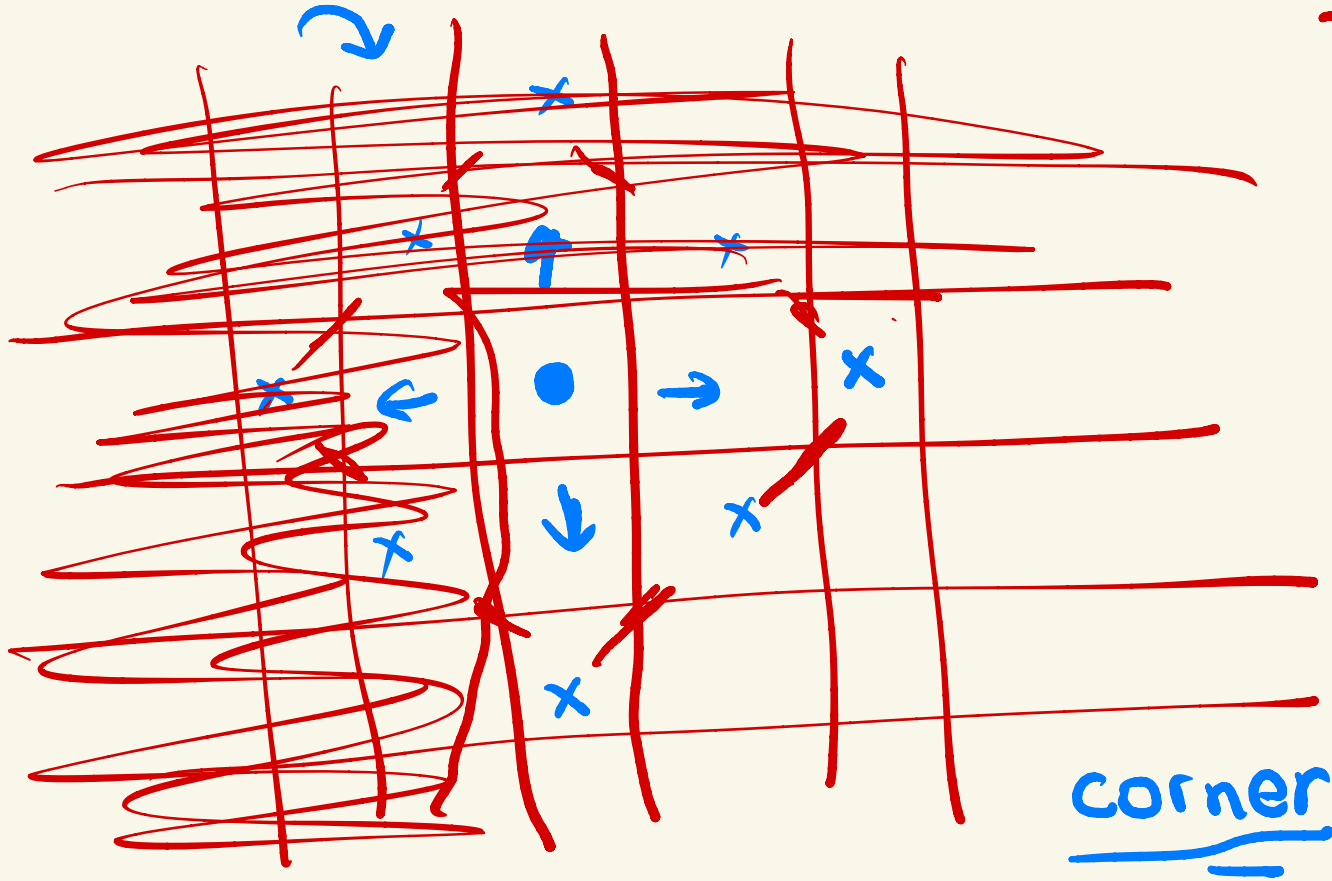
$$\begin{array}{r} 3, 3 = 3 \\ \hline \hookrightarrow 3 \end{array}$$

n rows

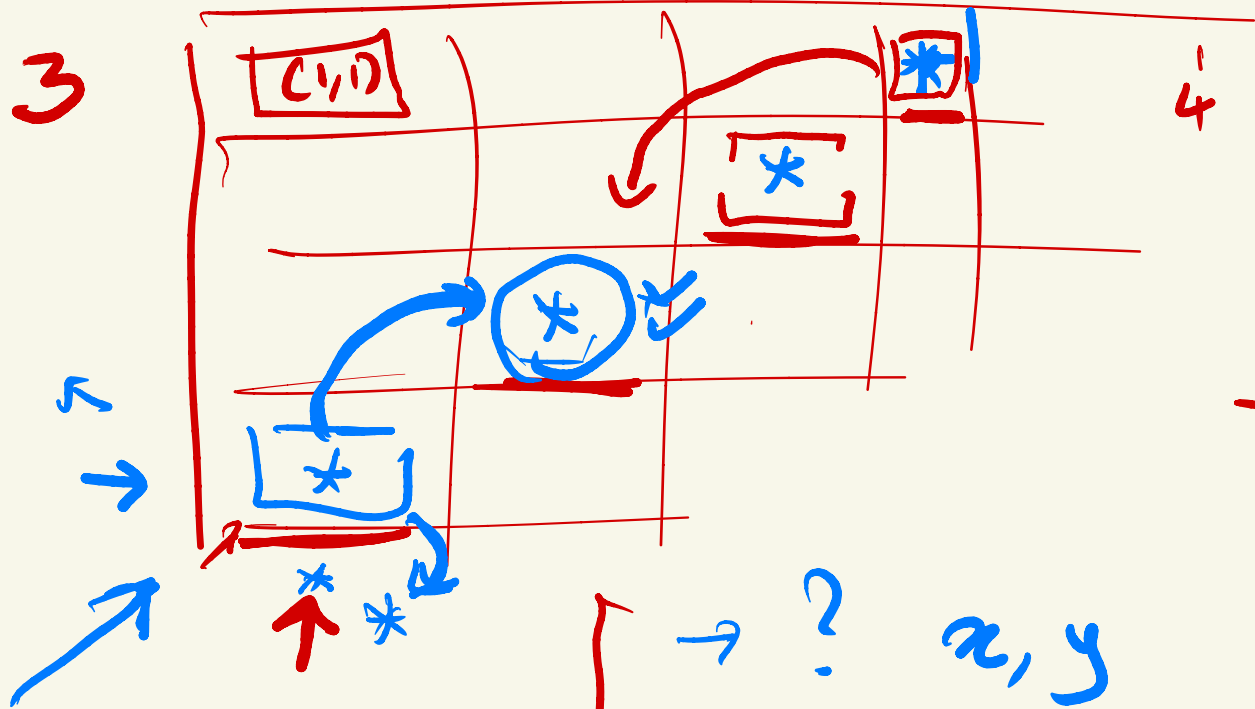
4 queries

First principles

2/1



3



4

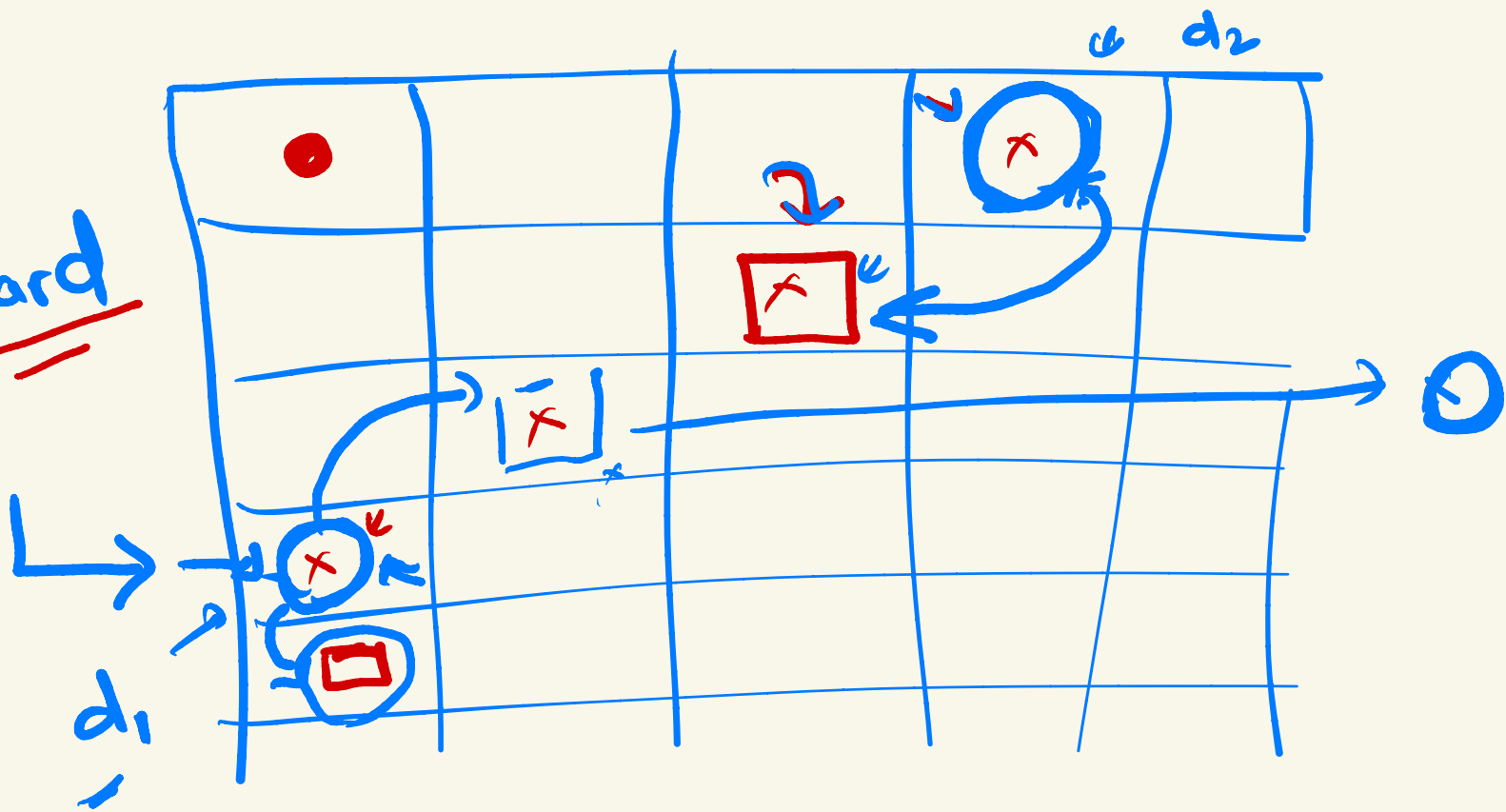
1 query

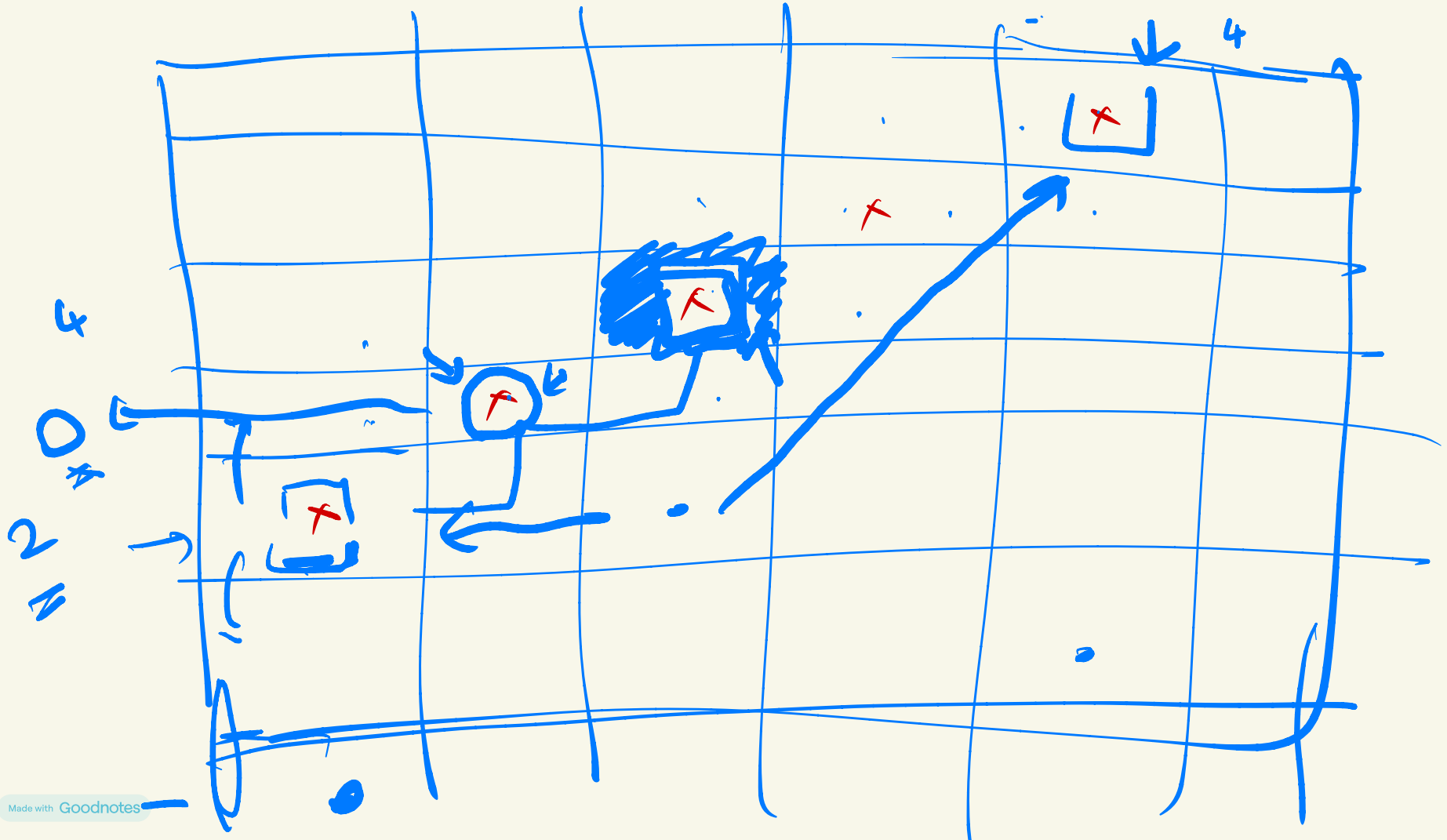
3 query

→ ? x, y
→ ? x, y

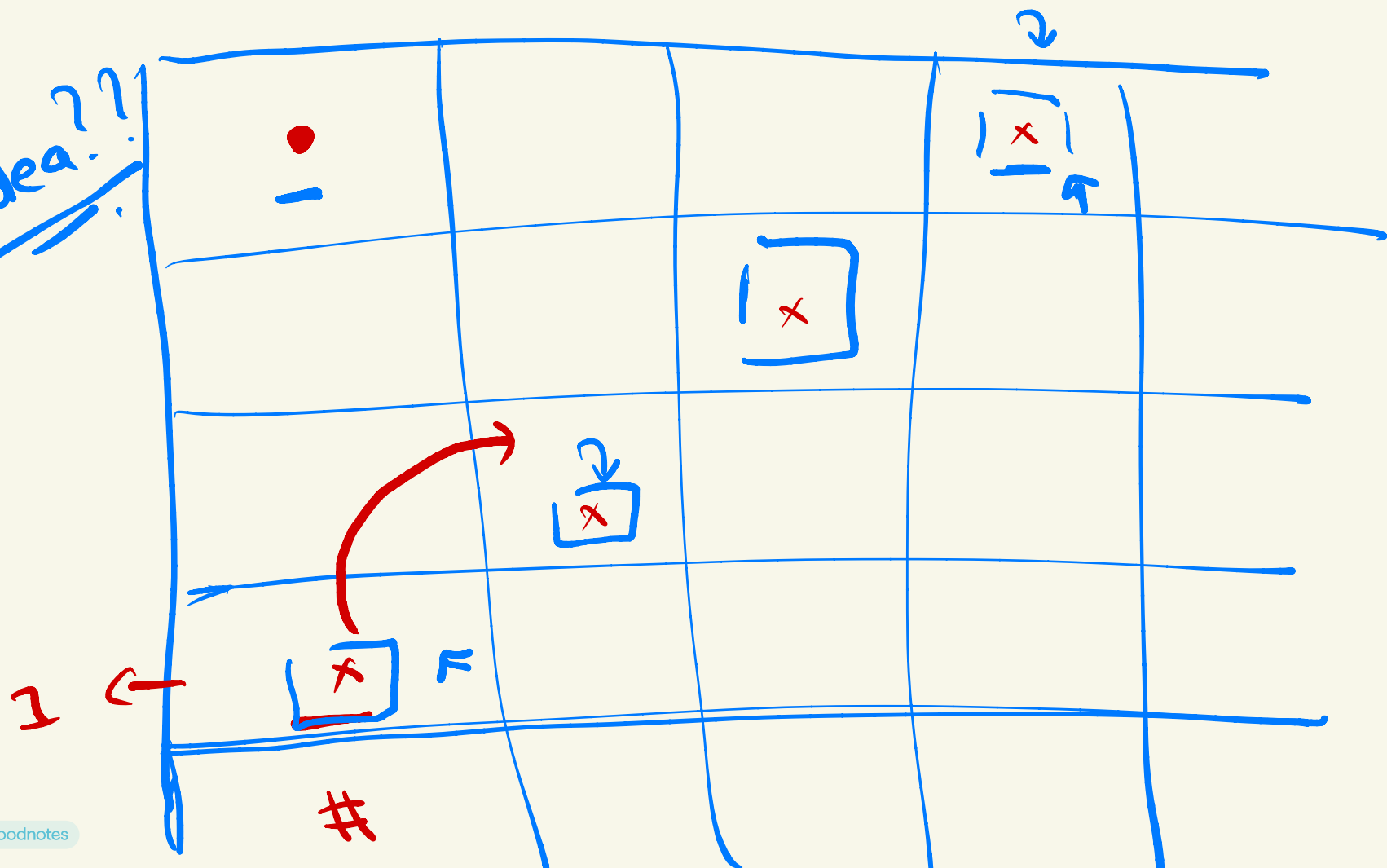
→ d → mine
x 0 x

Hard





idea??



$1, 1 \rightarrow x$

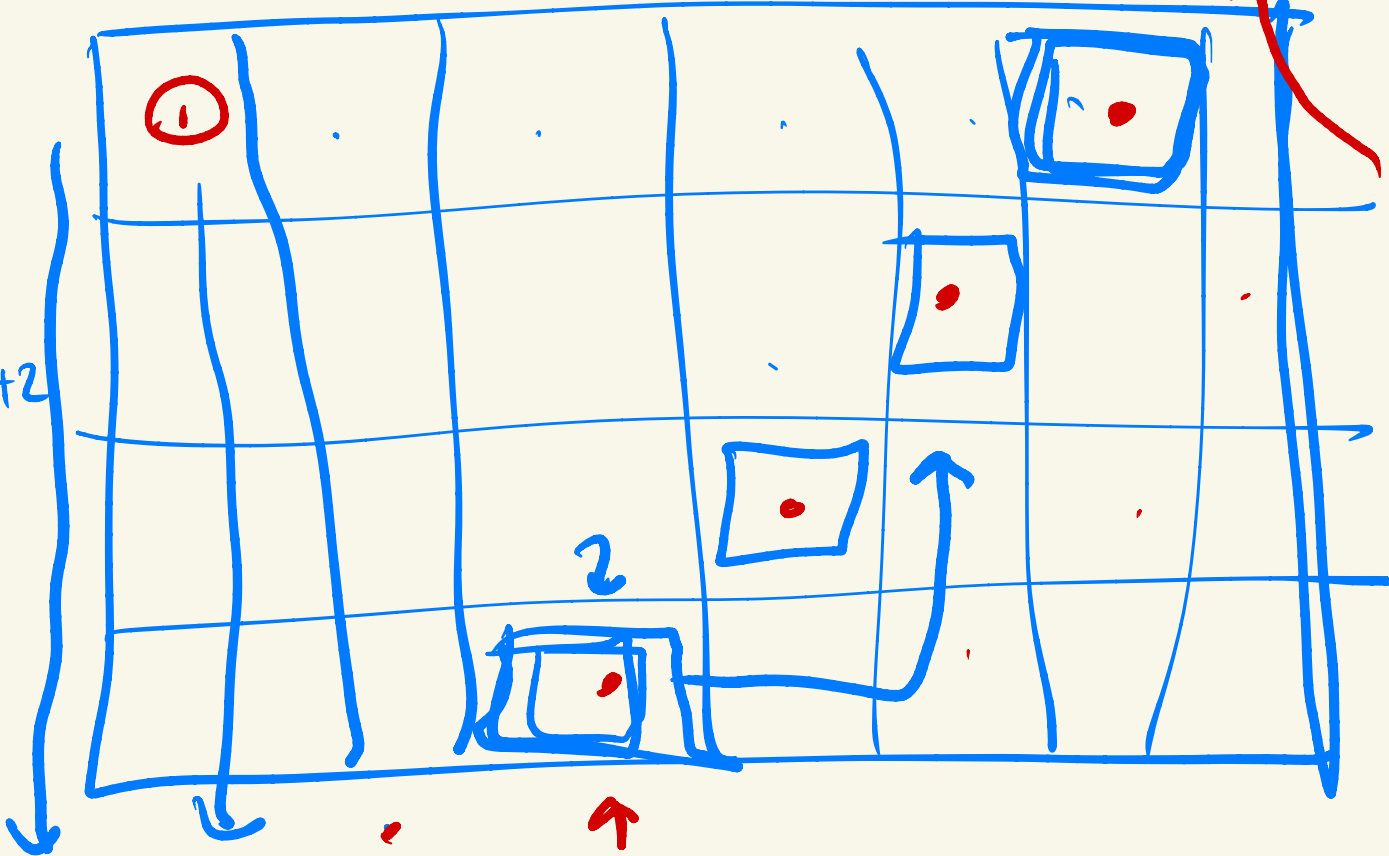
$1, 1 \rightarrow 1+x$
 $> m$

$(1+x-n)$

S

$4, 3$

$4, 3$



x_1 y_1

$$x_1 = \min(1 + d, n)$$

$$y_1 = d - x_1 + 1$$

query (x_1, y_1)

$\hookrightarrow d_1$

x_2

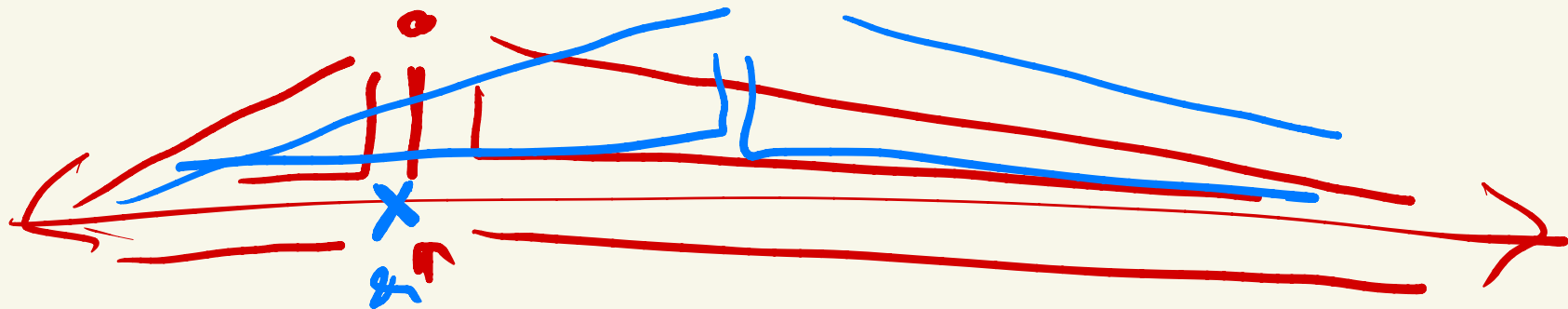
y_2

$$y_1 = \min(1 + d, m)$$

$$x_1 = d_1 - y_1 + 1$$

query (x_2, y_2)

d_2



✓✓ query(1,1) → d

x_1, y_1

$$\hookrightarrow \underline{x_1} = \min(n, 1+d)$$

$$\rightarrow y_1 = d - x_1 + 1$$

$$\rightarrow \text{query}(x_1, y_1) \quad d_1$$

x_2, y_2

$$\hookrightarrow y_2 = \min(m, 1+d)$$

$$x_2 = d - y_2 + 1$$

$$\rightarrow \text{query}(x_2, y_2) \quad d_2$$

d_i even $\text{query}(x_1 - d_{i/2}, y_1 + d_{i/2}) \rightarrow \min$

$$\times \quad \rightarrow x_2 + d_{i/2}, y_2 - d_{i/2}$$

Problem 3:

<https://codeforces.com/contest/1486/problem/C2>

C2. Guessing the Greatest (hard version)

time limit per test: 1 second

memory limit per test: 256 megabytes

The only difference between the easy and the hard version is the limit to the number of queries.

This is an interactive problem.



There is an array a of n different numbers. In one query you can ask the position of the second maximum element in a subsegment $a[l..r]$. Find the position of the maximum element in the array in no more than 20 queries.

A subsegment $a[l..r]$ is all the elements $a_l, a_{l+1}, \dots, a_r$. After asking this subsegment you will be given the position of the second maximum from this subsegment **in the whole** array.

Input

The first line contains a single integer n ($2 \leq n \leq 10^5$) — the number of elements in the array.

query (l, r)
↳ index of
second
maximum

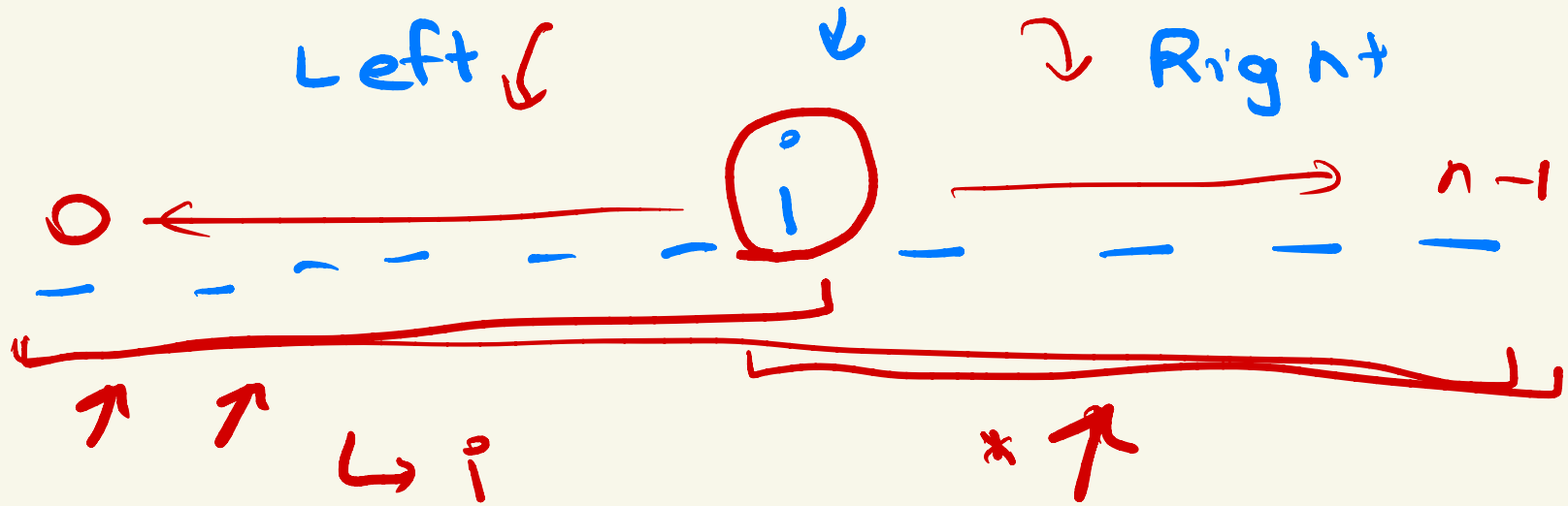
0, n-1

i
↓

10 1 2 3 5 8 7 6

↑ distinct

↓ second largest element



$O(\log N)$

query (i, j)

Binary searching
on j

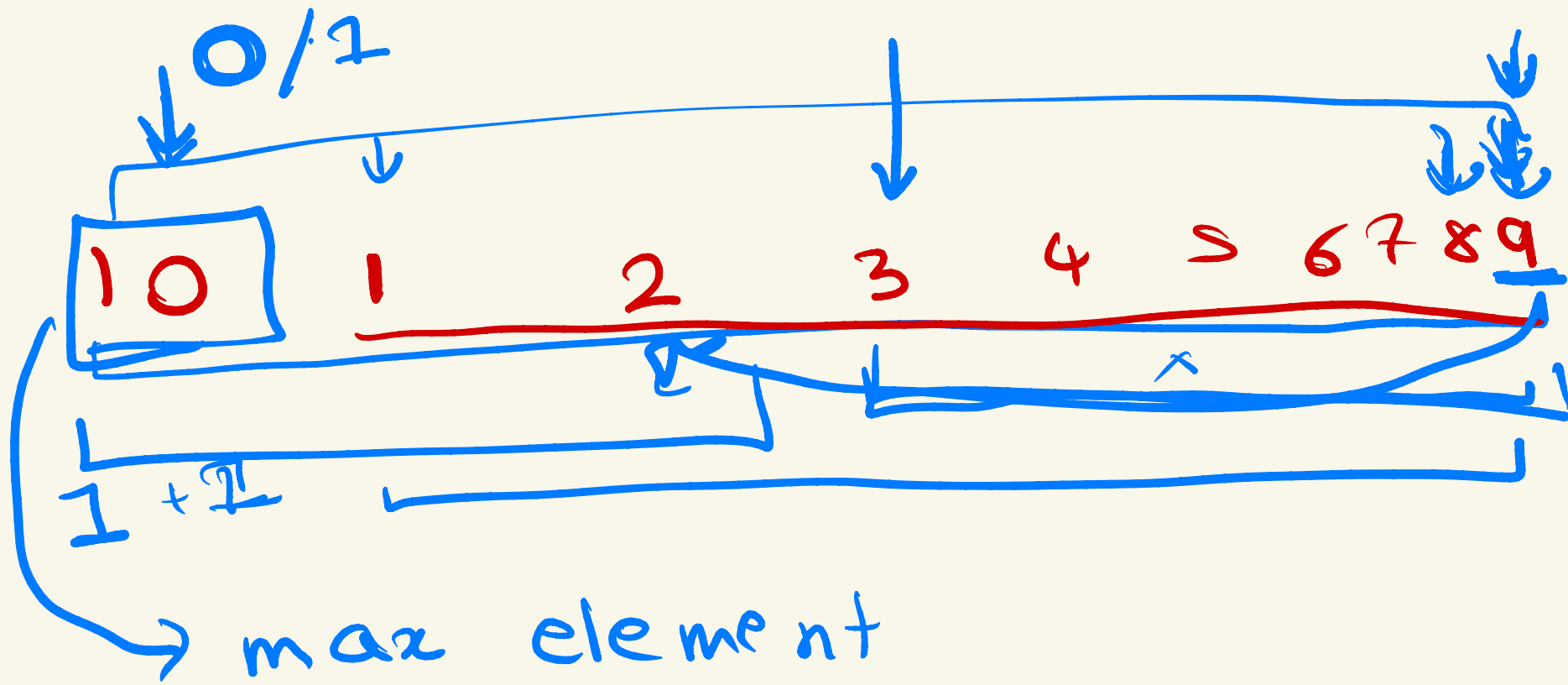
$$\underline{\underline{17}} + 3 = \underline{\underline{20}}$$

0
1
i

x n-1
query(i, j)

monotonic function

Binary Search



(x)

p

✓✓



o

(x)

